

Problem Set 7

The Net Change Theorem

September 18, 2026

Problem 5

(a)

$$\boxed{v(t) = t^2 + 3t - 4}$$

(b)

$$\begin{aligned}v(t) &= (t - 1)(t + 4) \\x(t) &= \frac{1}{3}t^3 + \frac{3}{2}t^2 - 4t + C \\\int_0^3 |v(t)| dt &= \int_1^3 v(t) dt - \int_0^1 v(t) dt \\&= x(3) - 2 \cdot x(1) \\&= 9 + 13.5 - 12 - 2\left(\frac{1}{3} + \frac{3}{2} - 4\right) \\&= \boxed{\frac{89}{6}m}\end{aligned}$$

Problem 7

(a)

max : 230

min : 172

The increasing condition is required, as the estimates are dependent on the left and right ends of the interval being the maximums and the minimums.

(b)

$$Q(6) = 6 + 17 + 30 + 41 + 50 + 57 = 201$$

Problem 8

(a)

$$A(t) = 200 + \int_0^t R(t) - D(t) dt$$

$$\begin{aligned}
&= 200 + \int_0^t 40 - t - \frac{t^2}{20} dt \\
&= 200 + \left(-\frac{t^3}{60} - \frac{t^2}{2} + 40t \right) \Big|_0^t \\
A(t) &= 200 + 40t - \frac{t^2}{2} - \frac{t^3}{60}
\end{aligned}$$

(b)

$$\begin{aligned}
A'(t) &= 40 - t - \frac{t^2}{20} \\
A'(10) &> 0 \\
A'(25) &< 0
\end{aligned}$$

The amount of water is increasing at $t = 10$, but is decreasing at $t = 25$.

(c)

$$\begin{aligned}
A'(t) = 0 &\implies 40 - t - \frac{t^2}{20} = 0 \\
t^2 + 20t - 800 &= 0 \\
(t + 40)(t - 20) &= 0 \implies t = 20 \\
A(20) &= 200 + 800 - 200 - \frac{400}{3} = \frac{2000}{3}
\end{aligned}$$

The amount of water is greatest at $t = 20$, with $\frac{2000}{3}$ liters of water.

(d)

$$t = 30 \implies A(30) = 500$$

The net change is not the same as the total amount of water entering, as the net change considers both the amount of water entering and leaving.

Problem 9

(a)

$$\begin{aligned}
P'(x) &= 3 - 0.01x + 0.000006x^2 \\
P(x) &= 3x - 0.005x^2 + 0.000002x^3 \\
&\int_{2000}^{4000} P'(x) dx \\
&= P(4000) - P(2000) \\
&= 60000 - 2000 \\
&= \boxed{58000}
\end{aligned}$$

(b)

$$\begin{aligned}
 P(t) &= 300 \sin\left(\frac{\pi t}{12}\right) \\
 \int_0^{12} P(t) dt &= -\frac{3600}{\pi} \cos\left(\frac{\pi t}{12}\right) \Big|_0^{12} \\
 &= \frac{7200}{\pi} \text{ Wh} \\
 &= \boxed{\frac{7.2}{\pi} \text{ kWh}}
 \end{aligned}$$

Problem 10

$$\begin{aligned}
 3 \int_0^a e^x dx &= \int_0^b e^x dx \\
 3e^a - 3 &= e^b - 1 \\
 3e^a - 2 &= e^b \\
 \boxed{b} &= \ln(3e^a - 2)
 \end{aligned}$$

Problem 11

(a)

$$\begin{aligned}
 D(T) &= \int_0^T v(t) dt = \int_0^T t^2 - 4t + 3 dt = \frac{T^3}{3} - 2T^2 + 3T \\
 D(T) &= \frac{T}{3}(T-3)^2 \\
 D(T) = 0 &\implies \boxed{T = 3}
 \end{aligned}$$

(b)

$$T \geq 0 \implies D(T) = \frac{T}{3}(T-3)^2 \geq 0$$

The particle is always at or to the right of its starting position.

(c)

$$\int_0^3 |v(t)| dt = \int_0^1 v(t) dt - \int_1^3 v(t) dt = 2 \cdot D(1) - D(3) = \boxed{\frac{8}{3}}$$

The total distance traveled is 8 meters. The particle can travel for the same distance in opposite directions to attain zero displacement.

(d)

$$\begin{aligned}\frac{8}{3} + \int_3^T v(t) dt &= 4 \\ \frac{T}{3}(T-3)^2 &= \frac{4}{3} \\ \boxed{T=4}\end{aligned}$$

(e)

$$\begin{aligned}v(t) &= (t-a)(t-b) = t^2 - (a+b)t + ab \\ D(T) &= \int_0^T v(t) dt = \int_0^T t^2 - (a+b)t + ab dt = \frac{T^3}{3} - \frac{(a+b)T^2}{2} + abT \\ D(b) &= \frac{b^3}{3} - \frac{ab^2 + b^3}{2} + ab^2 = \frac{b^2}{6}(3a - b) \\ D(b) = 0 &\implies \boxed{b = 3a} \\ \int_0^{3a} |v(t)| dt &= \int_0^a v(t) dt - \int_a^{3a} v(t) dt \\ &= 2 \cdot D(a) - D(3a) = 2 \left(\frac{a^3}{3} - 2a^3 + 3a^3 \right) = \boxed{\frac{8a^3}{3}}\end{aligned}$$